INCH-POUND

CCC-C-430G
15 September 2022
SUPERSEDING
CCC-C-430F
1 March 1984

FEDERAL SPECIFICATION

CLOTH, SHEETING, COTTON,0 AND POLYESTER AND COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a cotton sheeting, and polyester and cotton blend cloth.

….1.2 Classification. The cloth is available in the following types, classes and styles as specified.

1.2.1 Types.

Type I - Unbleached
Type II - Bleached or dyed

1.2.2 Classes.

Class 1 - Unshrunk
Class 2 - Shrunk

1.2.3 Styles.

Style A - Commercial designation no. 140 (muslin)
Style B - Commercial designation no. 128 (muslin)
Style C - Commercial designation no. 180 (percale)

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8305

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Style D - Polyester and cotton blend, 65/35 percent,
Commercial designation no. 180 (percale)
Style E - Polyester and cotton blend, 50/50 percent,
Commercial designation no. 180 (percale)
Style F - Polyester and cotton blend, 50/50 percent,
Commercial designation no. 120 (muslin)
Style G - Polyester and cotton blend, 65/35 percent,
Commercial designation no. 128 (muslin)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 - Rules and Regulations Under the Textile Fiber Products Identification Act
Title 16 CFR Part 1500 – Federal Hazardous Substances Act Regulations
Title 40, Protection of the Environment

(Copies of these documents are available online at https://ecfr.gov)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1	- Evaluation Procedure for Gray Scale for Color Change
AATCC EP2	- Evaluation Procedure for Gray Scale for Staining
AATCC EP8	- Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9	- Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8	- Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM16.3	- Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20	- Test Method for Fiber Analysis: Qualitative
AATCC TM20A	- Test Method for Fiber Analysis: Quantitative
AATCC TM61	- Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81	- Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM135	- Test Method for Dimensional Changes of Fabrics after Home

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737	- Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1424	- Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Yarn.

3.4.1.1 Styles A and B. The yarn for styles A and B shall be made of carded, drawn and spun 100 percent cotton.

3.4.1.2 Style C. The yarn for style C shall be made of carded, combed, drawn and spun 100 percent cotton.

3.4.1.3 Styles D and G. The yarn for styles D and G shall be a blend of polyester and cotton fiber, consisting of 65 (± 5) percent polyester and the remaining percentage cotton.

3.4.1.4 Styles E and F. The yarn for styles E and F shall be a blend of polyester and cotton fiber, consisting of 50 (± 5) percent polyester and the remaining percentage cotton.

3.4.2 Weave. The weave for the cloth shall be plain weave.

3.5 Color. The color of the finished dyed cloth shall be as specified in the contract or procurement documents (see 6.2).

3.5.1 Visual shade matching. The color and appearance of the dyed cloth shall match the standard sample when tested as specified in 4.6.1.

3.5.2 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The cloth shall be “Free” of labile sulfur when tested as specified in 4.6.2.

3.5.3 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished dyed cloth shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

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TABLE I. Colorfastness requirements (dyed cloth only).

	Laundering (Color Change and Staining) (3 cycles) (min.)	Light (after 40 AFU or 170 kJ/(m^2nm) @420 nm) (min.) 1/	Crocking (Dry and Wet) (min.)
Requirement	4	4	3.5

1/ AFU: AATCC Fading Units

3.6 Physical requirements. The finished cloth shall conform to the requirements, listed below, in Table II, when tested as specified in 4.5.

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TABLE II. Physical requirements.

Characteristic	Style A				Style B				Styles C, D and E	Styles F and G	Style G
	Type I		Type II		Type I		Type II		Type II	Type II	Type II
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Classes 1 and 2	Classes 1 and 2	Class 1
									Bleached only	Bleached only	Dyed only
Weight, oz./yd^2 (min.)	4.5	4.6	4.2	4.4	4.1	4.2	4.0	4.1	3.5	3.5	3.5
Total yarns per inch (min.)	140	140	140	140	128	128	128	128	180	128	128
Yarns per inch (min.)											
Warp	68	74	74	74	64	70	68	70	98	68	68
filling	72	69	66	69	64	63	60	63	82	60	60
Breaking strength, pounds, (min.)											
Warp	60	60	60	60	55	55	55	55	65	55	55
Filling	60	60	60	60	55	55	55	55	65	55	55
Tearing strength, pounds, (min.)											
Warp	-	-	-	-	-	-	-	-	-	-	6.0
Filling	-	-	-	-	-	-	-	-	-	-	3.8
Air permeability, ft^3/min/ft^2											
Minimum	-	-	-	-	-	-	-	-	-	-	120
Maximum	-	-	-	-	-	-	-	-	-	-	180

1/ A minus tolerance of 6 yarns per inch for the 128 count, 7 yarns per inch for the 140 count and 8 yarns per inch for the 180 count in either the warp and the filling directions is permitted, provided there is a total of not less than 128, 140 and 180 yarns per inch respectively.

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3.7 pH. The pH value of the water extract of the dyed and finished cloth shall be not less than 5.0 and not more than 8.5 when tested as specified in 4.5.

3.8 Dimensional change. The class 2 cloth shall not shrink or elongate more than 3.0 percent in either the warp and filling direction when tested as specified in 4.5. The stabilizing process shall not be identified.

3.9 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or procurement documents (see 6.2) and shall be the minimum acceptable width exclusive of the selvage.

3.10 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.11 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303, Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.6.3.

3.13 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance inspection (see 4.3.2).

4.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

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4.3.2 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract or purchase order, except where otherwise indicated (see 6.2).

4.4 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Type I of MIL-STD-3064.

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be two (2) continuous yards full width of the finished cloth for Class 1 and three (3) continuous yards full width for Class 2 cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. Material tests.

Characteristic	Requirement Reference	Test Method
Fiber identification	See 3.4.1	AATCC TM20 (see 6.5)
Fiber content	See 3.4.1	AATCC TM20A (see 6.5)
Weave	See 3.4.2	Visual
Color	See 3.5	See 4.6.1
Labile sulfur	See 3.5.2	See 4.6.2
Colorfastness: Laundering Light Crocking (wet and dry)	Table I	 AATCC TM61, Test 3A, 1/ AATCC TM16.3, Opt 3, 2/ AATCC TM8, 3/
Weight	Table II	ASTM D3776/D3776M, Opt. C
Total yarns per inch	Table II	ASTM D3775
Yarns per inch	Table II	ASTM D3775
Breaking strength	Table II	ASTM D5034
Tearing strength 4/	Table II	ASTM D1424
Air permeability 4/	Table II	ASTM D737
pH	See 3.7	AATCC TM81
Dimensional change	See 3.8	AATCC TM135 (3)(IV)(Aiii)

1/ Rated using the AATCC EP 1, and AATCC EP 2.
2/ Rated using the AATCC EP1.
3/ Rated using the AATCC EP8.
4/ Applicable to Type II, Class 1, Style G dyed cargo parachute canopy fabric only.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

4.6.2 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile

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sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as "pass" or "fail.

4.6.3 Toxicity. When toxicity requirement (see 3.12) can be demonstrated with historical use data on the finished treatments used, toxicity testing may not be required (see 6.2 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

a. Type I, Class 1, Style A, natural, for mattress, bed; and flags, ensign, guidons. (AS)
b. Type I, Class 2, Style A, for pillow covers.
c. Type I, Class 1, Style AB natural, for mattress covers. (N)
d. Type II, Class 1, Style A, white and gray-green, for pillowcases and bed sheets. (AS)
e. Type II, Class 1, Style B, white for bed sheets, men's and women's shirts. (N, AF)
f. Type II, Class 2, Style A, white, for coat, food handler's; mattress cover; hat, service trousers, men; uniform men. (N, MC)
g. Type II, Class 2, Style B, white, for cover, mattress; pad mattress, and stars for U.S. flags and Union Jack. (AS)
h. Type II, Class 1, Style C, white, for pillowcases and bed sheet. (VA)
i. Type II, Class 2, Style D and E, white, for pillowcases and bed sheets for Navy use. (N)

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j. Type I and II, Class 1, Style A, used by civil Government agencies for sheets, pillowcases, mattress and furniture covers, garment and laundry bags and ironing-board covers.
k. Type II, Class 1, Style G, dyed canopy fabric for the 12-foot cargo parachute which replaces cloth, muslin, cotton, Type III of MIL-C-4279.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class and style required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1)
e. Color of cloth required (see 3.5).
f. Width of cloth required (see 3.9).
g. Length required if other than specified (see 3.10).
h. When toxicity is required (see 3.12)
i. Inspection conditions (see 4.2).
j. Conformance inspection (see 4.3.2).
k. Packaging (see 5.1).

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens".

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for that specified requirement.

6.6 Toxicity testing. If dermal toxicity testing is required, it shall be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1
Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250)

6.7 Subject term (key word) listing.

Food handler coat
Mattress covers
Pillowcases
Sheets

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6.8 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY INTERESTS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FAS
DCG
VA-OSS
JUS-FPI

Custodians:
Army - GL
Navy – NU
Air Force - 11

Preparing Activity:
DLA-CT

(Project 8305-2022-019)

Review activities:
Army - EA, MD
Navy - MC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

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